

For example, in the case of the Cobb-Douglas production function (which can be written as $\ln Y = \beta_0 + \beta_K \ln K + \beta_L \ln L + u$) we can impose the restriction $\beta_K + \beta_L = 1$ for having constant returns to scale. Such linear restrictions for some scalar q and a K -dimensional vector r can be written as

$$H_0: r_1\beta_1 + \dots r_K\beta_K = r'\beta = q \quad \dots (7.20)$$

Here we can test the hypothesis that $r'\beta$ is BLUE for $r'\beta$. The variance is $V\{r'\beta\} = r'V\{\beta\}r$. Now if we replace σ^2 with its estimate s^2 in $V\{\beta\}$ we get estimated covariance matrix $\hat{V}\{\beta\}$. This gives us the standard error of the linear combination $r'\beta$ as $se(r'\beta) = s\sqrt{r'\hat{V}\{\beta\}r}$. As both β and $r'\beta$ follow normal distribution, we have

$$\frac{r'\beta - r'\beta}{se(r'\beta)} \sim t_{n-k} \quad \dots (7.21)$$

This is similar to (7.3.1). Hence, the test statistic can be written as

$$t_k = \frac{r'\beta - q}{se(r'\beta)} \quad \dots (7.22)$$

If the absolute value of t is greater than 1.96, it leads us to reject the null hypothesis at 5% level of significance.

7.4.1 Wald Test

In general, we come across the hypothesis which is a combination of the two null hypotheses and a set of J linear restriction on the coefficients of the model. The restriction in such cases is given as

$$R\beta = q \quad \dots (7.23)$$

where R is $J \times K$ matrix with an assumption that it is of full rank equal to the number of rows and q is a J -dimensional vector. For example, let us suppose a model with a set of restrictions

- $\beta_2 + \beta_3 + \dots + \beta_k = 1$
- $\beta_2 = \beta_3$.
- $J = 2$
- $R = \begin{bmatrix} 0 & 1 & 1 & \dots & \dots & 1 \\ 0 & 1 & -1 & 0 & \dots & 0 \end{bmatrix}$
- $q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Estimating the model with the above restrictions is possible by using (7.23). However, in some cases, estimating the model with these restrictions may not be possible under the null hypothesis. In such cases the result that

$$Rb \sim N(R\beta, \sigma^2, R(X'X)^{-1}R') \quad \dots (7.24)$$

can be used to formulate a quadratic form of equation.

It will be like (7.25) below which follows a Chi-squared distribution under the null hypothesis.

$$\xi = \frac{(Rb-q)'(R(X'X)^{-1}R')^{-1}(Rb-q)}{\sigma^2} \sim \chi_f^2 \quad \dots (7.25)$$

On replacing the unknown σ^2 in (7.25) by its estimate s^2 , we get a new statistic following a Chi-squared distribution. This statistic is referred to as the Wald Test. This test is based on the notion that the unrestricted model should 'approximately' satisfy the restrictions if they are true. Note that the Wald test provides the way for testing the restrictions without even estimating the restricted model.

7.4.2 Lagrange-Multiplier Test

Lagrange-Multiplier Test or the *LM*-test is used for testing the hypotheses expressed as one or more constraints on parameter values estimated under the likelihood framework. Unlike the Wald test, which requires estimation of the unrestricted model, *LM* -test requires estimation of the restricted parameter only. It is formulated to test whether the Lagrange multipliers which enforce the restrictions are significantly different from zero.

Let us suppose that $L(\theta)$ is a log-likelihood function of $(K \times 1)$ parameter vector θ and the score function and information matrix are given by

$$q(\theta) = \frac{\partial L(\theta)}{\partial \theta} \quad \dots (7.26)$$

$$I(\theta) = -E \left[\frac{\partial^2 L(\theta)}{\partial \theta \partial \theta'} \right] \quad \dots (7.27)$$

Let $\tilde{\theta}$ be the maximum likelihood estimator of θ subject to $(r \times 1)$ constraints $h(\theta) = \mathbf{0}$. Now the Lagrangian function is given by

$$\mathcal{L} = L(\theta) - \lambda' h(\theta) \quad \dots (7.28)$$

where λ is the $(r \times 1)$ vector of Lagrange multipliers. Now, the first-order conditions for $\tilde{\theta}$ are

$$\frac{\partial \mathcal{L}}{\partial \theta} = q(\tilde{\theta}) - H(\tilde{\theta})\tilde{\lambda} = \mathbf{0}$$

and

$$\frac{\partial \mathcal{L}}{\partial \lambda} = h(\tilde{\theta}) = \mathbf{0} \text{ where } H(\theta) = \partial h(\theta)' / \partial \theta.$$

Solving, we get the Lagrange Multiplier test statistic as

$$LM = \tilde{q}' \tilde{I}^{-1} \tilde{q} = \tilde{\lambda}' \tilde{H}' \tilde{I}^{-1} \tilde{H} \tilde{\lambda} \quad \dots (7.29)$$

where $\tilde{q} = q(\tilde{\theta})$, $\tilde{I} = I(\tilde{\theta})$ and $\tilde{H} = H(\tilde{\theta})$.

The term $\tilde{q}' \tilde{I}^{-1} \tilde{q}$ represents the 'score form' of the test statistic, whereas $\tilde{\lambda}' \tilde{H}' \tilde{I}^{-1} \tilde{H} \tilde{\lambda}$ represents the 'Lagrange Multiplier form' of the test statistic. In case we have an unrestricted MLE of θ , the score function $q(\theta) = \mathbf{0}$. However, it is not equal to zero in the case of restricted MLE($\tilde{\theta}$). If this constraint is true, we

would expect the values of $\tilde{\theta}$ and $\tilde{\lambda}$ to be small. Thus, large value of these two would lead to rejection of the null hypothesis $H_0: h(\theta) = \mathbf{0}$. If the constraint imposed under the null hypothesis is satisfied, the large sample LM statistic converges to chi-square distribution with $(k - r)$ degrees of freedom.

7.4.3 Likelihood Ratio Test

Likelihood ratio test or the LR -test enables us to test whether adding a set of new variables to the existing model increases the explaining capacity of the model. We saw that Wald test requires the estimation of only the unrestricted model, whereas, the LM -test requires the estimation of the restricted model. Unlike these two tests, the likelihood ratio test or LR -test requires estimation of both restricted as well as the unrestricted models.

Let us suppose that θ is a vector of parameter to be estimated having some restriction specified as the null hypothesis H_0 . The unrestricted and restricted maximum likelihood parameters of θ are denoted by $\hat{\theta}_u$ and $\hat{\theta}_R$ respectively. Under H_0 , the large sample distribution of $-2\ln\lambda$ is chi-square with degrees of freedom equivalent to the number of restrictions imposed, i.e., J . The LR test statistic is given by

$$LR = 2(\ln L(\hat{\theta}_u) - \ln L(\hat{\theta}_R)) \sim \chi_J^2 \quad \dots (7.30)$$

As said above, the LR -test enables us to test whether adding a set of new variables to the existing model increases the explaining capacity of the model. Using LR -test to test the simple null hypothesis against alternative hypothesis is not possible. This is because the degrees of freedom equals the number of parameters reduced as a consequence of imposing restrictions and in the case of testing the simple null hypothesis against alternative hypothesis, this would be equal to zero. Further, LR -test cannot be applied to test one distributional assumption against another as the likelihood functions of the two with different distributions will be unrelated.

7.4.4 Empirical Exercises

1) Suppose we are interested in finding out if the performance of a company, a crucial factor in determining the salary of the CEO. We have the model where the salary of the CEO is a function of the annual sales (*sales*), return on equity (*roe*), and return on stock (*ros*) as

$$\text{Salary} = \beta_1 + \beta_2 \text{sales} + \beta_3 \text{roe} + \beta_4 \text{ros} + \varepsilon_i$$

The following result is obtained:

$$\widehat{\text{salary}} = 4.3117 + 0.2803(\text{sales}) + 0.0174(\text{roe}) + 0.0002(\text{ros}) + u$$

(0.3154) (0.0353) (0.0041) (0.0005)

$$R^2 = 0.2826, \bar{R}^2 = 0.2721 \quad n = 209$$

(a) Provide the test for the significance of the explanatory variables (i.e., $H_0: \beta_2 = 0; \beta_3 = 0$ and $\beta_4 = 0$).

(a) To test the hypotheses $H_0: \beta_2 = 0; \beta_3 = 0$ and $\beta_4 = 0$, t statistic is calculated for each of the coefficients as follows.

(i) $H_0: \beta_2 = 0$

$$t_{sales} = \frac{b_k - \beta_k^0}{se(b_k)} = \frac{0.2803 - 0}{0.0353} = 7.9405$$

(ii) $H_0: \beta_3 = 0$

$$t_{roe} = \frac{b_k - \beta_k^0}{se(b_k)} = \frac{0.0174 - 0}{0.00409} = 4.2543$$

(iii) $H_0: \beta_4 = 0$

$$t_{ros} = \frac{b_k - \beta_k^0}{se(b_k)} = \frac{0.0002 - 0}{0.0005} = 0.4401$$

As the value for both t_{sales} and t_{roe} is greater than the critical value of 1.96, we reject the null hypothesis $H_0: \beta_2 = 0$ and $H_0: \beta_3 = 0$. Hence, we can say that both *sales* and *roe* are having positive influence on the salary of the CEO. The value for t_{ros} being small we cannot reject the null that $H_0: \beta_4 = 0$. Hence, it is statistically not significant.

(b) Test the overall significance of the model (i.e., $H_0: \beta_2 = \beta_3 = \beta_4 = 0$).

The overall significance of the model ($H_0: \beta_2 = \beta_3 = \beta_4 = 0$) can be tested by computing the F statistic

$$F = \frac{R^2/k - 1}{(1 - R^2)/(n - k)} = \frac{0.2826/(4 - 1)}{(1 - 0.2826)/(209 - 4)} = 26.91$$

Based on the large F statistic obtained, we get p -value equal to 0. This means we can reject the null hypothesis ($H_0: \beta_2 = \beta_3 = \beta_4 = 0$) that the performance of the company has no effect on the salary of the CEO at all significance level or that the overall model is significant.

(2) Suppose we have the following model

$$wage = \beta_1 + \beta_2 education + \beta_3 experience + \beta_4 tenure + \beta_5 age + \varepsilon_i$$

Now, we decide to drop *tenure* from the model as in many cases it would be equal to *experience*. Also, we decide in favour of dropping *age* as it may be highly correlated with *experience*. How would you test that dropping these two variables is acceptable?

Here, the null and the alternative is of the following nature

$$H_0: \beta_3 = \beta_4 = 0$$

$H_1: H_0$ is not true

The restricted model corresponding to H_0 will be

$$\text{wage} = \beta_1 + \beta_1 \text{education} + \beta_4 \text{experience} + u$$

$$\text{wage} = \beta_1 + \beta_2 \text{education} + \beta_4 \text{experience} + \varepsilon_i$$

To test the null, both the unrestricted and the restricted models are estimated. Based on the residual sum of the squares (RSS) obtained after estimation of the two models S_0 and S_1 respectively, the F statistic computed using

$$F = \frac{(RSS_0 - RSS_1)/J}{RSS_1/(n-k)}$$

Depending on the F statistic, p -value is determined. If we are able to reject the null hypothesis, we can conclude that *tenure* and *age* are jointly significant and hence dropping them is not acceptable. On the other hand, if we fail to reject the null, it would mean that they are not jointly significant and excluding the variable is acceptable.

Check Your Progress 3

- 1) Explain why we need to impose ‘linear restrictions’ in a regression model. Give an example.

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- 2) What is Wald’s test? On what rationale is it based? What is its advantage?

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- 3) In what way is the LM test different from the Wald’s test?

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- 4) What specific purpose does the LR Test serve? How is this test different from that of Wald’s and the LM tests?

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- 5) Why is it not possible to apply the LR test in the case of testing a simple null hypothesis against an alternative hypothesis?

7.5 LET US SUM UP

Evaluation of a regression model could be made through the coefficient of determination, which indicates the percentage variation in dependent variables that be explained by the independent variables. Hypothesis testing in multiple regression models is carried out to examine if a finding is sufficiently close to the stated hypothesis associated with the unknown population parameter vector β . After providing an outline of the coefficient of determination R^2 , details of the two types of tests. Commonly used tests, viz., the t -test and the F -Test are discussed in this unit. In practise, we come across hypotheses which require linear restrictions on more than one coefficient. Testing the hypotheses in such cases can be done either by reparameterization or by using alternative tests like the Wald, Lagrange Multiplier or the Likelihood ratio tests. These tests have been detailed out in the last section of the unit.

7.6 KEY WORDS

Lagrange Multiplier (LM) Test	: LM test is used for testing the hypotheses expressed as one or more constraints on the parameter values estimated under the likelihood framework.
Biased Estimator	: An estimator whose expectation, or sample mean, is different from the population value it is supposed to be estimating.
Likelihood Ratio Statistic	: A statistic that can be used to test single or multiple hypotheses when the constrained and unconstrained models are estimated by the maximum likelihood method.
Maximum Likelihood Estimation (MLE)	: A broadly applicable estimation method where the estimates are chosen to maximize the log-likelihood function.

- Quadratic Functions** : It refers to the functions that contain squares of one or more explanatory variables; they capture diminishing or increasing effects on the dependent variable.
- Restricted Model** : It is a model in hypothesis testing obtained after imposing all the restrictions required under the null.

7.7 SOME USEFUL BOOKS

- Greene, W. H., 2003, “*Econometric Analysis*.” Prentice Hall, New Jersey, USA. 5thed., pp. 484-496.
- Maddala, G. S., 1992, “*Introduction to Econometrics*”, Macmillan Publishing Company, New York. 2^d ed., pp. 127–177.
- Verbeek, M. 2004, “*A Guide to Modern Econometrics*”. John Wiley & Sons Ltd, England. 2nd ed., pp. 8–49.
- Wooldridge, Jeffrey M. 2000, “*Introductory Econometrics*”, South-Western College Publishing, 2000. Chapter 3; pp. 66–69.

7.8 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) It is defined as the ratio of Explained Sum of Squares (ESS) to Total Sum of Squares (TSS). Its expression is given by equation (7.3): $R^2 = \frac{ESS}{TSS} = \frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum(Y_i - \bar{Y})^2}$.
- 2) R^2 measures the proportion of the variation in Y which is explained by the multiple regression equation. Therefore, R^2 is often used as a goodness-of-fit statistic to compare the validity of regression results under alternative specifications of the independent variables in the model. Another reason is that the coefficient of determination (R^2) reflects the explanatory power of a regression model. A higher value of R^2 implies higher explanatory power of the underlying regression model compared to some other model. Thus, R^2 can be used as a measure of goodness of fit in multiple regressions.
- 3) $R^2 = 1 - \frac{\sum \hat{u}_i^2}{\sum(Y_i - \bar{Y})^2}$. This expression conveys that ‘as RSS decreases, R^2 increases’.
- 4) It is a fact that addition of a new explanatory variable invariably increases the value of R^2 . However, we cannot trust the inferences drawn from such models! This is because, there is a decrease in the degrees of freedom of RSS when we add explanatory variables to a model. A way out of this situation is to use adjusted- R^2 given by $\bar{R}^2 = 1 - (1 - R^2) \frac{(n-1)}{(n-k)}$ where n is number of observations in the regression model and k is number of

parameters in the equation. This approach eliminates the dependence of goodness of fit on the number of independent variables in the model.

- 5) One, when new variables are added to a regression model, while R^2 always increases, $\bar{R}^2$ may rise or fall. Second, The use of $\bar{R}^2$ eliminates the incentives to include numerous variables in a model. Third, $\bar{R}^2$ is always less than or equal to R^2 . Fourth, $\bar{R}^2$ can be negative also while R^2 is always positive. For these reasons, $\bar{R}^2$ is regarded as a more desirable goodness of fit measure compared to R^2 .

Check Your Progress 2

- 1) For testing $H_0: \beta_k = \beta_k^0$, we use the statistic $t_k = \frac{b_k - \beta_k^0}{se(\beta_k)}$. We reject the null hypothesis if the absolute value of the test statistic $|t_k|$ is very large. For large samples, at 5 % level of significance the null will be rejected if $|t_k| > 1.96$
- 2) The t -ratio refers to the t statistic that is computed to test the hypothesis that $\beta_k = 0$. Rejection of the null means that the estimate b_k differs significantly from 0 or that the corresponding regressor x_{ik} has significant impact on y_i .
- 3) A confidence interval is the interval consisting of all values for β_k^0 for which the t -test does not reject the null hypothesis that $\beta_k = \beta_k^0$. It provides a range of values for true β_k given the estimate b_k and its standard error. Thus, for the given probability $(1 - \alpha)$ it implies that $-t_{n-k; \alpha/2} < \frac{b_k - \beta_k^0}{se(b_k)} < t_{n-k; \alpha/2}$. If the level of significance is chosen as 5%, then the confidence interval implies that, in repeated sampling, in 95% cases, these intervals will contain true value of β_k .
- 4) The F -test is used to test for the joint significance of the estimated regression model. Specifically, it tests a hypothesis like: $H_0: \beta_2 = \beta_3 = \dots = \beta_K = 0$. Using the R^2 values of two regressions, viz., the full model denoted by RSS_I with that of the squared residuals of the restricted model (model with the last J regressor omitted) denoted by RSS_0 , the F -statistic is defined as: $F = \frac{(RSS_0 - RSS_I)/J}{RSS_I/(n-k)}$. In terms of R^2 , it is defined as: $\frac{R^2/(k-1)}{(1-R^2)/(n-k)}$.

Check Your Progress 3

- 1) The t -test considers restriction on one coefficient only like $\beta_k = 0$. In practise, we come across economic hypothesis which requires linear restrictions on more than one coefficients such as $\beta_2 + \beta_3 + \dots + \beta_k = 1$. For instance, in the case of the Cobb-Douglas production function (which is written in logarithmic form in a linear regression model), the constant returns to scale is represented by the sum of all the coefficients associated with the inputs as equal to one.
- 2) A statistic as in Equation (7.26), used to test J linear restrictions such as $H_0: R\beta = q$, is referred to as the Wald's test. It is based on the notion that

the unrestricted model should ‘approximately’ satisfy the restrictions if they are true. Its advantage is that it provides a method for testing the restrictions without requiring the estimation of the restricted model.

- 3) Wald’s test requires the estimation of the ‘unrestricted model’. It does not require the estimation of the ‘restricted model’. The LM test, on the other hand, requires the estimation of the ‘restricted parameter’ only.
- 4) Likelihood ratio test or the *LR*-test enables us to test whether adding a set of new variables to the existing model increases the explaining capacity of the model. Unlike the Wald’s and the LM test, which requires the estimation of either the restricted or the unrestricted models, the likelihood ratio test requires the estimation of both the restricted as well as the unrestricted models.
- 5) This is because in the LR test, the degree of freedom equals the number of parameters reduced as a consequence of imposing the restrictions. In the case of testing the simple null hypothesis against alternative hypothesis, this would be equal to zero.



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